

MASENO UNIVERSITY
MMA 115 BASIC MATHEMATICS
2024/2025 ACADEMIC YEAR

QUESTION ONE (COMPULSORY)(30 marks)

- (a) (i) Find the sum to infinity of the geometric series $16 + 12 + 9 + \dots$ [3 marks]
(ii) The sum of three consecutive terms of an arithmetic progression is 36 and their product is 1428. Find the three terms given that the common difference is non positive. [3 marks]
- (b) (i) An arc subtends an angle of 0.25 radian at the center of a circle, and a sector of area 18cm^2 is bounded by the arc and the two radii. Find the radius of the circle. [3 marks]
(ii) Solve the equation $\sin^2 x + \sin x \cos x = 0$, for $0 \leq x \leq 360$. [3 marks]
(iii) Find the maximum value of $1 - 2 \cos \theta$ hence state the smallest value of θ , $0 \leq x \leq 360$ for which the value occurs. ✓ [3 marks]
- (c) (i) Evaluate
(I) 6C_4
(II) 7P_3 [4 marks]
(ii) Write $7 + 2i$ in polar coordinate form. [3 marks]
(iii) Find the remainder when $8x^3 + 4x^2 + 6x + 7$ is divided by $2x + 1$. [2 marks]
- (d) Solve for x to 3 decimal places $\log_3 x + \log_x 9 = 3$. ✓ [3 marks]
- (e) Given the sets $T = \{a, c, d, e\}$ and $S = \{b, c, d, f\}$, find $T \Delta S$ [3 marks]

QUESTION TWO [20 marks]

- (a) Given that α and β are acute angles with $\sin \alpha = \frac{7}{25}$ and $\cos \beta = \frac{5}{13}$, find without using tables or a calculator
(i) $\cos(\alpha + \beta)$ [4 marks]
(ii) $\tan(\alpha + \beta)$ [3 marks]
- (b) Prove that $\csc 2\theta + \cot 2\theta = \cot \theta$ [5 marks]
- (c) The polynomial $6x^3 + 7x^2 + ax + b$ has a remainder of 72 when divided by $(x - 2)$ and is exactly divisible by $(x + 1)$. Find a and b . Show that $(2x - 1)$ is also a factor of the polynomial. [5 marks]
- (d) Solve for x in the equation $(0.8)^{2x} = 0.45$. [3 marks]

QUESTION THREE [20 marks]

- (a) Expand $(1 + x)^5$ and use the expansion to find;
(i) $(1.01)^5$

(ii) $(0.2)^5$. [6 marks]

(b) In how many ways can 4 red, 3 yellow and 2 green discs be arranged in a row if disks of the same color are not to follow each other? [4 marks]

(c) Four men Peter, Roger, Smith and Thomas are to be seated at a circular table. In how many ways can this be done so that Roger sit next to Smith? [3 marks]

(d) (i) Simplify $\frac{5+2i}{1-3i}$. [4 marks]

(ii) Expand and simplify $(\cos 30 + i \sin 30)^2$. [3 marks]

QUESTION FOUR [20 marks]

(a) Prove that the sum of the series $1^2 + 2^2 + 3^2 + \dots + n^2$ is $\frac{n}{6}(n+1)(2n+1)$ [10 marks].

(b) A and B are two points on the circumference of a circle center O and radius 5cm. Angle $AOB = \theta$ radians and length of chord $AB = 6\text{cm}$. Find θ and area of the minor segment bounded by AB and the circle. [5 marks]

(c) Find the sum of the arithmetic progression $8.5 + 12 + 15.5 + 19 + \dots + 103$. [5 marks]

QUESTION FIVE [20 marks]

(a) The following shows results of a survey on the types of exercise taken by a group of 100 people.

Run (R) 65

Swim (S) 48

Cycle (C) 60

Run and Swim 40

Swim and Cycle 30

Run and Cycle 35

Do all the three 25

(i) Draw a venn diagram to represent the data above. [5 marks]

(ii) Find the probability that none take the exercise. [2 marks]

(iii) Find the probability that someone swims but does not run. [2 marks]

(iv) Find the probability that a person does at least two of these types of exercise. [3 marks]

(v) Jasper is one of the person in the group. Given that Jasper runs, find the probability that he swims but does not cycle. [2 marks]

(b) List the elements of the set $A = \{x^2 : x \in \mathbb{Z}^+ \text{ and } x < 6\}$ [2 marks]

(c) Given that $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ is a universal set, $Q = \{3, 5, 7\}$ and $R = \{2, 4, 6, 8\}$; Find [2 marks]

(i) Q' [2 marks]

(ii) $Q \cap R$